

Part 2

Random Variables, Probability Distributions

Binomial Distribution

Normal Distribution, Standard Normal Distribution

The Central Limit Theorem

Poisson Distribution

Population versus Sample

Population – The set of all objects under consideration. Objects in a population are called *elements*.

Examples

- Boxes of a particular brand of corn flakes.
- Newly made light bulbs in a warehouse at a given time.

Another Definition of Population

- All possible outcomes of an *experiment* can also be values in a population.
- Flipping a coin twice. Possible outcomes are: HH, HT, TH, TT. Could make the outcomes numeric using 3, 2, 1, 0 to be the elements corresponding to the above *events*.
- If the experiment is repeated many times, the resulting values constitute a population: 2, 0, 3, 1, 0, 2, ...

Random Sampling

- **Sample** – A subset of the elements from a population.
- **Simple Random Sample** of size n from a **finite population** is a subset selected in such a way that every subset of n elements is equally likely to be the one selected. (Abbreviation: SRS)
- A random sample from a small finite population can be obtained using the **Random Numbers** - using code or random number table.
 - `sample(1:100,3)` or `sample(3)`
- Computer programs are used to draw samples from large populations.

Probability, Random Variables, and Probability Distributions

- Assume a numeric finite population of N elements, and suppose n_1 of the elements are the number k .
- Suppose a SRS of size 1 is taken from this population.
- The **probability** that the number k is selected is n_1/N .
- That is, we define this probability to be the relative frequency of occurrence of k in the population.

Probability: Example

- Example: $N = 10$
- Population: 1, 1, 1, 3, 7, 7, 9, 9, 9, 9
- Denote by $P(k)$, the probability of the number k being selected
- $P(1) = 3/10$, $P(3) = 1/10$, $P(7) = 2/10$,
 $P(9) = 4/10$, $P(2) = 0/10$, $P(1 \text{ or } 3) = 4/10$,
 $P(\text{not } 7) = 1 - 2/10$.

Random Variables

A **random variable** is a function defined on a numeric population. We will often say that this function value is **the value assumed by the random variable**.

- **Example:** $N = 10$
- **Population:** 1, 1, 1, 3, 7, 7, 9, 9, 9, 9
- Let X be a random variable defined on this population.
- X can assume any of the values 1, 3, 7, or 9.

Random Variables (continued)

- A **Random variables** are different from ordinary variables because they assume values from the population with associated probabilities.
- For e.g., we will say
$$P(X = 1) = 3/10, \quad P(X = 7) = 2/10,$$
$$P(X = 2) = 0/10, \quad P(X = -6) = 0/10,$$
$$P(1 \leq X \leq 3) = 4/10, \quad P(1 \leq X \leq 9) = 1. \text{ etc.}$$
- A **probability distribution** specifies the probabilities associated with all possible values the random variable X can assume.

The M & M Example

Example:

If X represents the number of RED *M&Ms* in a vending machine size package, then perhaps the probability distribution of X is something like the following:

x	5	6	7	8	9	10	11
$P(X = x)$	$\frac{2}{28}$	$\frac{2}{28}$	$\frac{5}{28}$	$\frac{7}{28}$	$\frac{5}{28}$	$\frac{4}{28}$	$\frac{3}{28}$

The M & M Example (continued)

We read this table by finding a value of X , say 8, then reading the probability below it:

- The probability of observing 8 M&Ms in a package is $\frac{7}{28}$, meaning, if we open many many packages of M&Ms, about 7 out of 28 will have exactly 8 M&Ms within.

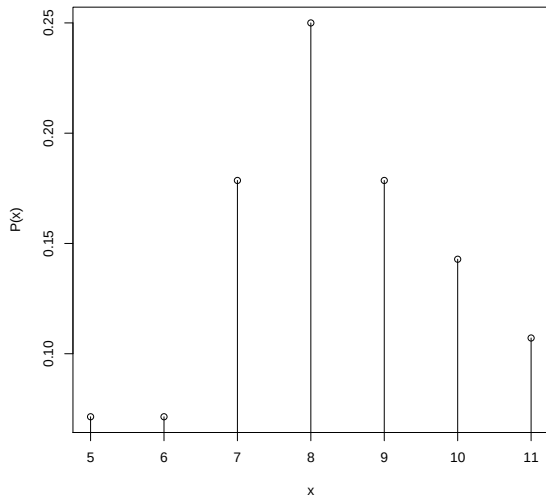
This is an example of a **discrete random variable**.

Discrete Random Variables

- When the elements of a population consists of a countable discrete set of values the random variable defined on that population is called **discrete random variable**.
- For discrete random variables the probability distribution can be displayed using a histogram or a table of values and associated probabilities as shown above.

The M & M Probability Distribution

Probability Distribution for M & M Data



Continuous Random Variables

- When the population consists of an uncountable number of values in a line interval use integration to calculate proportions of the population that exist in any given **subinterval** of a continuum.
- For a random variable defined on such a population we will deal with the probability that the random variable assumes a value in a given interval, not that it assumes a specific numeric value.

Continuous Random Variables (contd.)

- When the population consists of a set of values in a continuous range the random variable defined on that population is called **continuous random variable**.
- The probability distribution of a continuous random variable is a smooth curve.
- The **area under the curve** between specified values of the random variable correspond to a probability.
- The probability that a continuous random variable assumes a specific numeric value is zero.

Continuous Random Variables (contd.)

- **Example:** A Continuous Random Variable X defined on
- a **Population:** all numbers in the interval $(0,2)$ occurring with various frequencies (that we won't specify here).
- By Definition
$$P(1/2 \leq X \leq 1/2) \equiv P(X = 1/2) = 0$$
$$P(0 \leq X \leq 2) = 1, \quad P(X < 0) = P(X \leq 0) = 0$$
- We could possibly calculate the following probability
$$P(1 \leq X \leq 2) = ?$$